

# On collapsibility and confounding bias in Cox and Aalen regression models

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- **Prioritizing Bias Control**

When variables are limited, we fear estimation direction errors (Bias) more than lack of precision (Efficiency).

- **Population-level Impact**

Public health decisions focus on the effect of implementing a policy across the whole population (Marginal Effect), not just subgroup associations.

# The data

$$T_m(t) = g(\lambda(t \mid x = 1), \lambda(t \mid x = 0)).$$

$T_m(t)$ : compare treated vs untreated *without controlling*  $Z$  (mix all  $Z$  levels).

$$T(t) = g(\lambda(t \mid \hat{x} = 1), \lambda(t \mid \hat{x} = 0)).$$

$T(t)$ : compare two hypothetical worlds: set everyone to  $X = 1$  vs set everyone to  $X = 0$ .

where  $\hat{x} = x$  is shorthand for the intervention  $do(X = x)$ .

$$T_c(t) = g(\lambda(t \mid x = 1, Z), \lambda(t \mid x = 0, Z)).$$

$T_c(t)$ : compare treated vs untreated *within the same  $Z$  level* (or after adjusting for  $Z$ ).

# The Assumption

## Confounding bias at time $t$ :

$$\Delta(t) = T(t) - T_m(t).$$

- $T(t)$  is the causal marginal effect (*what would happen under intervention*).
- $T_m(t)$  is the marginal association (*what we see without adjustment*).

To identify  $T(t)$  from observational data, we **assume  $Z$  is sufficient to adjust for confounding**, so that  $\lambda(t \mid \hat{x} = x, Z) = \lambda(t \mid X = x, Z)$ . After controlling  $Z$ , treated and untreated are comparable.

# The Trap and The Fix

## Problem in Cox models

Even when  $Z$  is *not* a confounder (e.g.,  $Z$  independent of  $X$ ), the Cox hazard ratio can change after adjusting for  $Z$ .

This is **non-collapsibility**.

So a change in hazard ratio after adjustment does not automatically imply confounding.

# The Trap and The Fix

The common change-in-estimate idea compares:

$$T_c(t) - T_m(t).$$

But in Cox settings, this difference generally mixes two components:

$$T_c(t) - T_m(t) = \underbrace{(T(t) - T_m(t))}_{\text{confounding bias } \Delta(t)} + \underbrace{(T_c(t) - T(t))}_{\text{non-collapsibility component}}.$$

- $\Delta(t)$ : what we would like to call “confounding bias”
- $T_c(t) - T(t)$ : difference between conditional effect and marginal causal effect

# The Trap and The Fix

Goal: estimate the marginal causal quantities under interventions:

G-computation idea

$$P(T|\hat{x} = x) = \int P(T|X = x, Z = z) dF_Z(z),$$

# G-computation formula

The G-computation formula gives the distribution of the survival time  $T$  under intervention:

$$P(T|\hat{x} = x) = \int P(T|X = x, Z = z) dF_Z(z),$$

where  $F_Z$  is the marginal distribution of confounder  $Z$ .

For comparison, the observed distribution of the survival time  $T$  conditioned on exposure  $X$  is

$$P(T|X = x) = \int P(T|X = x, Z = z) dF_{Z|X=x}(z).$$

If  $Z$  is a confounder, then  $F_Z(z) \neq F_{Z|X=x}(z)$ .

# Causal survival function

Using the G-computation formula, the causal survival function is

$$\begin{aligned} S(t|\hat{x} = x) &= P(T > t|\hat{x} = x) = \int P(T > t|X = x, Z = z)dF_Z(z) \\ &= \int S(t|X = x, Z = z)dF_Z(z) \\ &= \int e^{-\Lambda(t|x, z)}dF_Z(z). \end{aligned}$$

Here,

$$\Lambda(t|x, z) = \int_0^t \lambda(s|x, z)ds$$

is the cumulative conditional hazard function.

# Causal hazard function

The hazard function satisfies  $\lambda(t) = -\frac{d}{dt}\{\log S(t)\}$ .

Using the causal survival function, the causal hazard function is

$$\begin{aligned}\lambda(t|\hat{x} = x) &= -\frac{d}{dt}\{\log S(t|\hat{x} = x)\} \\ &= -\frac{d}{dt} \log \left( \int e^{-\Lambda(t|x,z)} dF_z(z) \right) \\ &= -\frac{\frac{d}{dt} \int e^{-\Lambda(t|x,z)} dF_z(z)}{\int e^{-\Lambda(t|x,z)} dF_z(z)} \\ &= \frac{\int e^{-\Lambda(t|x,z)} \lambda(t|x,z) dF_z(z)}{\int e^{-\Lambda(t|x,z)} dF_z(z)}.\end{aligned}$$

# Aalen additive hazards model

The Aalen additive hazards model is defined as

$$\lambda(t|x, z) = \beta_0(t) + \beta_X(t)x + \beta_Z(t)z.$$

The corresponding cumulative hazard function is

$$\Lambda(t|x, z) = B_0(t) + B_X(t)x + B_Z(t)z.$$

We will use the risk difference as an association measure:  $g(a, b) = a - b$ .

# Marginal and causal survival function

The survival function under Aalen's model is

$$\begin{aligned} S(t|x, z) &= e^{-\Lambda(t|x, z)} \\ &= e^{-B_0(t) - B_X(t)x - B_Z(t)z}. \end{aligned}$$

The marginal survival function observed under this model is

$$\begin{aligned} S(t|X = x) &= \int S(t|x, z) dF_{Z|X=x}(z) \\ &= E[S(t|x, z)|X = x] \\ &= E[e^{-B_0(t) - B_X(t)X - B_Z(t)Z}|X = x] \\ &= e^{-B_0(t) - B_X(t)x} E[e^{-B_Z(t)Z}|X = x]. \end{aligned}$$

In contrast, the causal survival function is

$$S(t|\hat{x} = x) = E[S(t|x, z)].$$

# Marginal and causal hazard function

The hazard function satisfies  $\lambda(t) = -\frac{d}{dt}\{\log S(t)\}$ .

Using the marginal survival function, the marginal hazard function is

$$\begin{aligned}\lambda(t|X=x) &= -\frac{d}{dt} \log \left( e^{-B_0(t)-B_X(t)x} E[e^{-B_Z(t)Z}|X=x] \right) \\ &= -\frac{d}{dt} (-B_0(t) - B_X(t)x + \log E[e^{-B_Z(t)Z}|X=x]) \\ &= \beta_0(t) + \beta_X(t)x + \beta_Z(t) \frac{E[e^{-B_Z(t)Z}Z|X=x]}{E[e^{-B_Z(t)Z}|X=x]}.\end{aligned}$$

The causal hazard function is

$$\lambda(t|\hat{x}=x) = \beta_0(t) + \beta_X(t)x + \beta_Z(t) \frac{E[e^{-B_Z(t)Z}Z]}{E[e^{-B_Z(t)Z}]}.$$

# Exposure effects under Aalen's model

So far, we obtained by definition of Aalen's model

$$\lambda(t|x, z) = \beta_0(t) + \beta_X(t)x + \beta_Z(t)z.$$

Therefore, the conditional exposure effect is

$$\mathcal{T}_c(t) = \lambda(t|X = 1, Z) - \lambda(t|X = 0, Z) = \beta_X(t).$$

Using the G-computation formula, we obtained the causal hazard function

$$\lambda(t|\hat{x} = x) = \beta_0(t) + \beta_X(t)x + \beta_Z(t) \frac{E[e^{-B_Z(t)Z}Z]}{E[e^{-B_Z(t)Z}]},$$

such that the marginal exposure effect (causal effect) is

$$\mathcal{T}(t) = \lambda(t|\hat{x} = 1) - \lambda(t|\hat{x} = 0) = \beta_X(t).$$

This shows that there is collapsibility for Aalen's model.

# Marginal association

We also obtained the marginal hazard function

$$\lambda(t|X = x) = \beta_0(t) + \beta_X(t)x + \beta_Z(t) \frac{E[e^{-B_Z(t)Z}Z|X = x]}{E[e^{-B_Z(t)Z}|X = x]}.$$

Therefore, the marginal association is

$$\begin{aligned}\mathcal{T}_m(t) &= \lambda(t|X = 1) - \lambda(t|X = 0) \\ &= \beta_X(t) + \beta_Z(t) \left( \frac{E[e^{-B_Z(t)Z}Z|X = 1]}{E[e^{-B_Z(t)Z}|X = 1]} - \frac{E[e^{-B_Z(t)Z}Z|X = 0]}{E[e^{-B_Z(t)Z}|X = 0]} \right)\end{aligned}$$

Thus,  $\mathcal{T}(t) \neq \mathcal{T}_m(t)$  in general when  $Z$  is a confounder, showing that there is a confounding bias.

# Amount of confounding bias

Normally,

$$\text{Confounding bias} = \mathcal{T}(t) - \mathcal{T}_m(t)$$

However, we want to compute the confounding bias on the cumulative scale. Therefore,

$$\begin{aligned}\text{Confounding bias} &= \int_0^t (\mathcal{T}(s) - \mathcal{T}_m(s)) ds \\ &= B_X(t) - B_X^m(t),\end{aligned}$$

where

$$\beta_X^m(t) = \beta_X(t) + \beta_Z(t) \left( \frac{E[e^{-B_Z(t)Z} Z | X = 1]}{E[e^{-B_Z(t)Z} | X = 1]} - \frac{E[e^{-B_Z(t)Z} Z | X = 0]}{E[e^{-B_Z(t)Z} | X = 0]} \right).$$

# Cox proportional hazards model

- Conditional hazard function:

$$\lambda(t|x, z) = \lambda_0(t)e^{\beta_x x + \beta_z z}$$

where  $\lambda_0(t)$  denotes the baseline hazard function.

- Cumulative conditional hazard function:

$$\begin{aligned}\Lambda(t|x, z) &= \int_0^t \lambda(u|x, z) du \\ &= \int_0^t \lambda_0(u) e^{\beta_x x + \beta_z z} du \\ &= e^{\beta_x x + \beta_z z} \int_0^t \lambda_0(u) du \\ &= \Lambda_0(t) e^{\beta_x x + \beta_z z}\end{aligned}$$

where  $\Lambda_0(t) = \int_0^t \lambda_0(u) du$ .

# Cox proportional hazards model

- Cumulative conditional hazard function:

$$\Lambda(t|x, z) = \Lambda_0(t)e^{\beta_X x + \beta_Z z}$$

- Conditional survival function:

$$\begin{aligned} S(t|x, z) &= \exp(-\Lambda(t|x, z)) \\ &= e^{-\Lambda_0(t) \exp(\beta_X x + \beta_Z z)} \end{aligned}$$

Define

$$S_0\{x, \theta(t); z\} := e^{-\Lambda_0(t) \exp(\beta_X x + \beta_Z z)}$$

as the individual survival function, where  $\theta(t) = \{\beta_X, \beta_Z, \Lambda_0(t)\}^T$ .

# Marginal hazard function

$$\lambda(t|x) = \lambda_0(t)e^{\beta_X x} j\{x, \theta(t)\}$$

where

$$j\{x, \theta(t)\} = \frac{E[S_1\{x, \theta(t); Z\}|X = x]}{E[S_0\{x, \theta(t); Z\}|X = x]}$$

with

$$S_0\{x, \theta(t); z\} = e^{-\Lambda_0(t) \exp(\beta_X x + \beta_Z z)},$$

$$S_1\{x, \theta(t); z\} = S_0\{x, \theta(t); z\} e^{\beta_Z z},$$

$$\text{and } \theta(t) = \{\beta_X, \beta_Z, \Lambda_0(t)\}^T.$$

# Marginal hazard function

$$\begin{aligned}\lambda(t|x) &= \frac{\int e^{-\Lambda(t|x,z)} \lambda(t|x,z) dF_{Z|X=x}(z)}{\int e^{-\Lambda(t|x,z)} dF_{Z|X=x}(z)} \\&= \frac{E[S_0\{x, \theta(t); z\} \lambda_0(t) e^{\beta_{xx} + \beta_{zz}} | X = x]}{E[S_0\{x, \theta(t); z\} | X = x]} \\&= \lambda_0(t) e^{\beta_{xx}} \frac{E[S_0\{x, \theta(t); z\} e^{\beta_{zz}} | X = x]}{E[S_0\{x, \theta(t); z\} | X = x]} \\&= \lambda_0(t) e^{\beta_{xx}} \frac{E[S_1\{x, \theta(t); z\} | X = x]}{E[S_0\{x, \theta(t); z\} | X = x]} \\&= \lambda_0(t) e^{\beta_{xx}} j\{x, \theta(t)\}\end{aligned}$$

where

$$\begin{aligned}S_1\{x, \theta(t); z\} &= S_0\{x, \theta(t); z\} e^{\beta_{zz}}, \\j\{x, \theta(t)\} &= \frac{E[S_1\{x, \theta(t); Z\} | X = x]}{E[S_0\{x, \theta(t); Z\} | X = x]}\end{aligned}$$

# Causal hazard function

$$\lambda(t|\hat{x} = x) = \lambda_0(t)e^{\beta_{xx}}h\{x, \theta(t)\}$$

where

$$h\{x, \theta(t)\} = \frac{E[S_1\{x, \theta(t); Z\}]}{E[S_0\{x, \theta(t); Z\}]}$$

The derivation is as follows:

$$\begin{aligned}\lambda(t|\hat{x} = x) &= \frac{\int e^{-\Lambda(t|x,z)}\lambda(t|x,z)dF_Z(z)}{\int e^{-\Lambda(t|x,z)}dF_Z(z)} \\ &= \lambda_0(t)e^{\beta_{xx}}\frac{E[e^{\beta_{zz}}S_0\{x, \theta(t); z\}]}{E[S_0\{x, \theta(t); z\}]} \\ &= \lambda_0(t)e^{\beta_{xx}}\frac{E[S_1\{x, \theta(t); z\}]}{E[S_0\{x, \theta(t); z\}]} \\ &= \lambda_0(t)e^{\beta_{xx}}h\{x, \theta(t)\}\end{aligned}$$

The marginal exposure effect (casual effect) is given by

$$\begin{aligned}\mathcal{T}(t) &= g(\lambda(t|\hat{x} = 1), \lambda(t|\hat{x} = 0)) \\ &= \log \frac{\lambda(t|\hat{x} = 1)}{\lambda(t|\hat{x} = 0)} \\ &= \log \frac{\lambda_0(t)e^{\beta_X h\{1, \theta(t)\}}}{\lambda_0(t)h\{0, \theta(t)\}} \\ &= \beta_X + \log \left[ \frac{h\{1, \theta(t)\}}{h\{0, \theta(t)\}} \right]\end{aligned}$$

so, in general,  $\mathcal{T}(t) \neq \beta_X$  even when  $Z$  and  $X$  are independent, and therefore non-collapsibility is present.

The marginal association is

$$\begin{aligned}\mathcal{T}_m(t) &= g(\lambda(t|x=1), \lambda(t|x=0)) \\ &= \log \frac{\lambda(t|x=1)}{\lambda(t|x=0)} \\ &= \log \frac{\lambda_0(t)e^{\beta_X j\{1, \theta(t)\}}}{\lambda_0(t)j\{0, \theta(t)\}} \\ &= \beta_X + \log \left[ \frac{j\{1, \theta(t)\}}{j\{0, \theta(t)\}} \right]\end{aligned}$$

It again follows that confounding is present,  $\mathcal{T}(t) \neq \mathcal{T}_m(t)$ , whenever  $\beta_X \neq 0$ , and  $Z$  and  $X$  are not independent.

# Confounding Bias

Because of the non-collapsibility, the Cox model may not hold for both the conditional and marginal relationship, which further emphasizes that naïve measures of confounding can be misleading.

Let

$$\hat{h}\{x, \theta(t)\} = \frac{\sum_{i=1}^n S_1\{x, \theta(t); Z_i\}}{\sum_{i=1}^n S_0\{x, \theta(t); Z_i\}},$$
$$\hat{j}\{x, \theta(t)\} = \frac{\sum_{i=1}^n S_1\{x, \theta(t); Z_i\} I(X_i = x)}{\sum_{i=1}^n S_0\{x, \theta(t); Z_i\} I(X_i = x)}$$

denote the empirical estimates. It then follows that the amount of confounding bias can be estimated by

$$\begin{aligned}\hat{\Delta}(t) &= \hat{\mathcal{T}}(t) - \hat{\mathcal{T}}_m(t) \\ &= \log \left( \frac{\hat{h}\{1, \hat{\theta}(t)\}}{\hat{h}\{0, \hat{\theta}(t)\}} \right) - \log \left( \frac{\hat{j}\{1, \hat{\theta}(t)\}}{\hat{j}\{0, \hat{\theta}(t)\}} \right)\end{aligned}$$

# Large Sample Properties - Aalen Model

- Recall that the Estimated Confounding Bias

$$\Delta \hat{B}(t) = \hat{B}_X(t) - \hat{B}_X^m(t)$$

- convergence to Gaussian process

$$n^{1/2}[\hat{B}_X(t) - \hat{B}_X^m(t) - \{B_X(t) - B_X^m(t)\}] = n^{-1/2} \sum_{i=1}^n \epsilon_i^B(t) + o_p(1),$$

where  $\epsilon_i^B(t)$  are zero mean i.i.d. terms.

$$\epsilon_i^B(t) = \int_0^t AR^{-1}(s)Y_i(s)dM_i(s) - \int_0^t A^m\{R^m\}^{-1}(s)Y_i^m(s)dM_i^m(s)$$

By CLT,

$$n^{-1/2} \sum_{i=1}^n \epsilon_i^B(t) = n^{1/2} \left( \frac{1}{n} \sum_{i=1}^n \epsilon_i^B(t) \right) \xrightarrow{d} \mathcal{GP}(0, \Sigma(t))$$

# Large Sample Properties - Cox Model

For cox model, three estimators are proved to converge to Gaussian process.

- Confounding Bias:

$$n^{1/2}\{\hat{\Delta}(t) - \Delta(t)\} = n^{-1/2} \sum_{i=1}^n \epsilon_i^{\Delta}(t) + o_p(1)$$

- Casual Effect:

$$n^{1/2}\{\hat{\mathcal{T}}(t) - \mathcal{T}(t)\} = n^{-1/2} \sum_{i=1}^n \epsilon_i^{\mathcal{T}}(t) + o_p(1)$$

- Amount of non-linearity:

$$n^{1/2}\{\hat{\Delta}_{nl}(t) - \Delta_{nl}(t)\} = n^{-1/2} \sum_{i=1}^n \epsilon_i^{\Delta_{nl}}(t) + o_p(1)$$

where  $\epsilon_i^{\Delta}(t), \epsilon_i^{\mathcal{T}}(t), \epsilon_i^{\Delta_{nl}}(t)$  are all zero mean i.i.d. terms.

And the variance of these three Gaussian models are  $n^{-1} \sum_{i=1}^n \hat{\epsilon}_i^{\Delta}(t)^2$ ,  $n^{-1} \sum_{i=1}^n \hat{\epsilon}_i^{\mathcal{T}}(t)^2$  and  $\hat{\Sigma}(t) = n^{-1} \sum_{i=1}^n \hat{\epsilon}_i^{\Delta_{nl}}(t)^2$  respectively.

## Simulation Parameters:

- **Confounder Z:** Continuous variable,  $Z \sim N(0, 0.5^2)$ .
- **Exposure X:** Binary, generated as  $P(X = 1|Z = z) = \text{logit}^{-1}(0.3z)$ .
- **True Causal Effect:**  $\beta_X = \log(2)$ .
- **Confounding Strength:**  $\beta_Z = \log(2)$ .
- **Baseline Hazard ( $\lambda_0$ ):** Two scenarios tested:
  - **High Risk:**  $\lambda_0 = 0.1$ .
  - **Low Risk:**  $\lambda_0 = 0.1/8 = 0.0125$ .

# Three Measures of Confounding Bias

- We evaluate the performance of our method using three metrics:

1. True Bias ( $\Delta(t)$ ): The theoretical gold standard.

$$\Delta(t) = \mathcal{T}(t) - \mathcal{T}_m(t)$$

*Difference between the causal effect and the marginal association.*

2. Aalen Estimator ( $\hat{\Delta}(t)$ ): The proposed linear approach.

$$\hat{\Delta}(t) = \hat{B}_X(t) - \hat{B}_X^m(t)$$

*Uses the difference in Aalen coefficients to isolate pure confounding.*

3. Naive Estimator ( $\hat{\Delta}_{Si}$ ): The traditional Cox approach.

$$\hat{\Delta}_{Si} = \hat{\beta}_X - \hat{\beta}_X^m$$

*Simple difference between adjusted and unadjusted Cox hazard ratios.*

# Result Analysis: Visual Comparison (Figure 1)

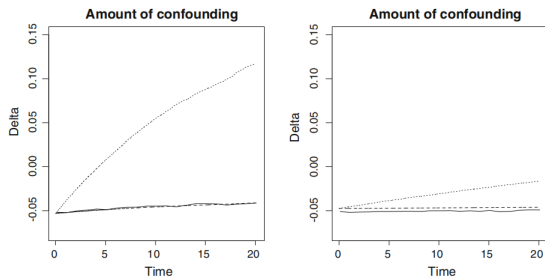


Figure: Figure 1

Left display:  $\lambda_0 = 0.1$ . Right display:  
 $\lambda_0 = 0.1/8$ .

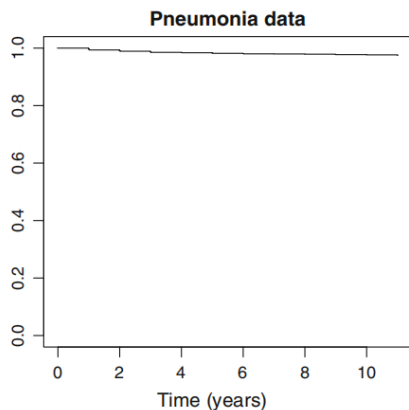
## Key Observations from Fig 1:

- **Aalen Estimator  $\hat{\Delta}(t)$ :** The dashed line tracks the solid (true) line well in both scenarios. This confirms it as an **\*\*unbiased\*\*** measure.
- **Naive Cox  $\hat{\Delta}_{Sj}$ :** The dotted line deviates severely from the truth when the hazard is **\*\*High\*\*** x

# Result Analysis: Table 1

| $(n, \lambda_0(t))$ |                                    | $t = 5$ | $t = 10$ | $t = 15$ |
|---------------------|------------------------------------|---------|----------|----------|
| (100,0.1/8)         | Mean $\hat{\Delta}(t) - \Delta(t)$ | 0.001   | -0.001   | -0.002   |
|                     | SSE( $\hat{\Delta}(t)$ )           | 0.087   | 0.083    | 0.081    |
|                     | SEE( $\hat{\Delta}(t)$ )           | 0.092   | 0.087    | 0.086    |
|                     | 95 % CP( $\hat{\Delta}(t)$ )       | 0.959   | 0.965    | 0.964    |
| (200,0.1/8)         | Mean $\hat{\Delta}(t) - \Delta(t)$ | -0.004  | -0.006   | -0.008   |
|                     | SSE( $\hat{\Delta}(t)$ )           | 0.059   | 0.057    | 0.055    |
|                     | SEE( $\hat{\Delta}(t)$ )           | 0.059   | 0.057    | 0.056    |
|                     | 95 % CP( $\hat{\Delta}(t)$ )       | 0.936   | 0.943    | 0.954    |
| (100,0.1)           | Mean $\hat{\Delta}(t) - \Delta(t)$ | 0.000   | 0.001    | 0.002    |
|                     | SSE( $\hat{\Delta}(t)$ )           | 0.070   | 0.073    | 0.080    |
|                     | SEE( $\hat{\Delta}(t)$ )           | 0.072   | 0.073    | 0.078    |
|                     | 95 % CP( $\hat{\Delta}(t)$ )       | 0.961   | 0.961    | 0.961    |
| (200,0.1)           | Mean $\hat{\Delta}(t) - \Delta(t)$ | -0.001  | -0.001   | -0.001   |
|                     | SSE( $\hat{\Delta}(t)$ )           | 0.059   | 0.058    | 0.057    |
|                     | SEE( $\hat{\Delta}(t)$ )           | 0.058   | 0.059    | 0.065    |
|                     | 95 % CP( $\hat{\Delta}(t)$ )       | 0.937   | 0.940    | 0.951    |

# Case Study: Pneumonia Data (Low Risk)



## Variables Definition:

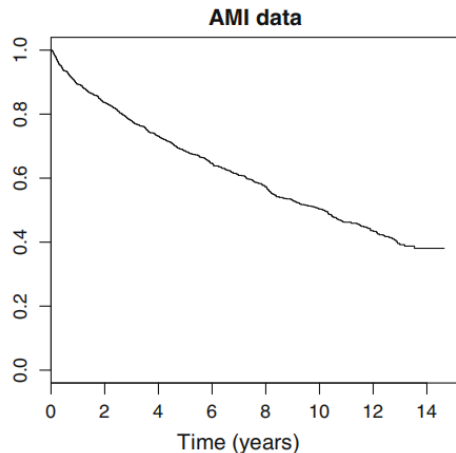
- **X (Exposure):** Breast feeding (1=yes, 0=no)
- **Y (Outcome):** Time to hospitalized pneumonia in 1st year
- **Z (Confounder):** Maternal smoking

**Figure:** Kaplan Meier Curve for Pneumonia Dataset

## Key Results & Conclusions:

- **Rare Phenomenon:** Only 73 events out of 3,470 infants;  $\lambda_0(t) \approx 0.0025$ .
- **Model Consistency:** Causal effect  $\mathcal{T}(t)$  and conditional Cox estimates are **almost identical**.
- **Takeaway:** In low-risk settings, Cox model's non-linearity is not an issue.

## Case 2: AMI data (High Risk)



### Variables Definition:

- **X (Exposure):**  
Aspirin(1=yes,0=no)
- **Y (Outcome):** Time to death from all causes
- **Z (Confounder):** Age

### Data Distribution (Table 2):

|              | Age (years) |         |         |         |
|--------------|-------------|---------|---------|---------|
|              | 24 – 53     | 54 – 61 | 62 – 70 | 71 – 95 |
| Aspirin, no  | 53          | 62      | 81      | 110     |
| Aspirin, yes | 214         | 190     | 170     | 140     |

Figure: Kaplan Meier Curve for AMI Dataset

# AMI Results: Confounding and Non-linearity

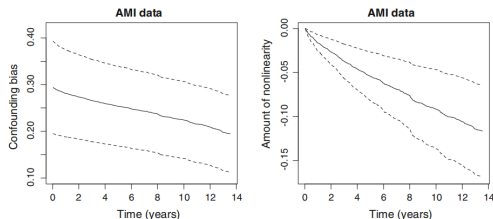


Figure: Figure 3: Bias analysis in AMI data

Left: Estimated confounding bias  $\hat{\Delta}(t)$ .  
Right: Amount of non-linearity  $\hat{\beta}_c - \hat{T}(t)$ .

## Interpretation:

- **Confounding Bias:** The left plot shows a bias of  $\sim 0.25$ , confirming that age strongly confounds the aspirin effect.
- **Cox Failure:** There is a noticeable difference between the conditional Cox estimate and the true causal effect  $\hat{T}(t)$ .
- **Conclusion:** In high-risk AMI patients, Cox models produce biased results due to model non-linearity.

# Multiple Confounders

## Overall Confounding Bias vs. Bias Due to Covariate $Z$

|                   | Overall Confounding Bias                                           | Bias Due to Single Variable $Z$                                                                           |
|-------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Research Question | How large is the bias if no covariates are adjusted?               | How much confounding can be ascribed to one of the covariates?                                            |
| Comparison Target | Causal effect $\mathcal{T}$ & Marginal association $\mathcal{T}_m$ | Effect adjusted only for $W$ ( $\mathcal{T}_Z$ ) & Effect adjusted for both $W$ and $Z$ ( $\mathcal{T}$ ) |
| Formula           | $\mathcal{T}(t) - \mathcal{T}_M(t)$                                | $\mathcal{T}_Z(t) - \mathcal{T}(t)$                                                                       |

# Multiple Confounders - Formula Details

We divide all covariates into  $V = (Z, W)$ .

Confounding bias due to a scalar covariate  $Z$  is

$$\mathcal{T}_Z(t) - \mathcal{T}(t)$$

- $\mathcal{T}_Z(t) = g(\lambda_Z(t|\hat{X} = 1), \lambda_Z(t|\hat{X} = 0))$
- $\mathcal{T}(t) = g(\lambda(t|X = 1), \lambda(t|X = 0))$

$\lambda_Z(t|\hat{X} = x)$  can be obtained by only doing G-computation to  $W$ .

$$P_Z(T|\hat{X} = x) = \int P(T|X = x, W = w) dF_W(w).$$

# Multiple Confounders - Calculating on Aalen Model

First, we got

$$\lambda(t|x, z, w) = \beta_0(t) + \beta_X(t)x + \beta_W(t)w + \beta_Z(t)z.$$

Goal: Obtain  $\lambda_Z(t|\hat{X} = x)$

Derive in four steps:

- 1  $\lambda(t|x, z, w) \Rightarrow \lambda(t|x, w)$ : Marginalizing out  $Z$
- 2  $\lambda(t|x, w) \Rightarrow S(t|x, w)$ : Hazard to Survival,  $S(t) = \exp(-\int_0^t \lambda(s)ds)$
- 3  $S(t|x, w) \Rightarrow S_Z(t|\hat{X} = x)$ : G-computation over  $W$
- 4  $S_Z(t|\hat{X} = x) \Rightarrow \lambda_Z(t|\hat{X} = x)$ : Survival to Hazard,  $\lambda(t) = -\frac{d}{dt} \log S(t)$

# Multiple Confounders - Calculating on Aalen Model

Finally,

$$\lambda_Z(t|\hat{X} = x) = \beta_0(t) + \beta_X(t)x + h(t; x),$$

where

$$h(t; x) = \frac{\int e^{-B_W(t)w - B_Z^*(t; B_Z, x, w)} \{ \beta_W(t)w + \beta_Z(t)l(B_Z(t); x, w) \} dF_W(w)}{\int e^{-B_W(t)w - B_Z^*(t; B_Z, x, w)} dF_W(w)}$$

with

$$l(B_Z(t); x, w) = \frac{E(e^{-B_Z(t)Z} Z | X = x, W = w)}{E(e^{-B_Z(t)Z} | X = x, W = w)},$$
$$B_Z^*(t; B_Z, x, w) = \int_0^t l(B_Z(s); x, w) dB_Z(s).$$

# Multiple Confounders - Calculating on Aalen Model

With

- $\lambda(t|x, z, w) = \beta_0(t) + \beta_X(t)x + \beta_W(t)w + \beta_Z(t)z$
- $\lambda_Z(t|\hat{X} = x) = \beta_0(t) + \beta_X(t)x + h(t; x)$

The corresponding association measure is

- $\mathcal{T}(t) = \lambda(t|1, z, w) - \lambda(t|0, z, w) = \beta_X(t)$
- $\mathcal{T}_Z(t) = \lambda_Z(t|\hat{X} = 1) - \lambda_Z(t|\hat{X} = 0) = \beta_X(t) + h(t; 1) - h(t; 0)$

The amount of confounding bias due to  $Z$  is then

$$\Delta_Z(t) = H(t; 1) - H(t; 0) = \int_0^t [h(s; 1) - h(s; 0)] ds$$

# Multiple Confounders - Problems in Empirical Estimator

The theoretical confounding bias is shown above, while in practice, we use the empirical estimator

$$\begin{aligned}\hat{H}(t; x) = & \int_0^t \frac{\int e^{-\hat{B}_W(t)w - B_Z^*(t; \hat{B}_Z, x, w)} w d\hat{F}_W(w)}{\int e^{-\hat{B}_W(t)w - B_Z^*(t; \hat{B}_Z, x, w)} d\hat{F}_W(w)} d\hat{B}_W(s) \\ & - \int_0^t \frac{\int e^{-\hat{B}_W(t)w - B_Z^*(t; \hat{B}_Z, x, w)} I(\hat{B}_Z(t); x, w) d\hat{F}_W(w)}{\int e^{-\hat{B}_W(t)w - B_Z^*(t; \hat{B}_Z, x, w)} d\hat{F}_W(w)} d\hat{B}_Z(s)\end{aligned}$$

In this "terrible" formula,  $\hat{B}_W(t)$  and  $\hat{B}_Z(t)$  are coefficients of Aalen model, and  $\hat{F}_W(w)$  is the empirical distribution function of  $W$ .

Only  $I(\hat{B}_Z(t); x, w)$  is unobtainable from the sample data.

# Multiple Confounders: Solutions for $I$

To calculate  $I(\hat{B}_Z(t); x, w)$ , the conditional distribution  $Z \mid V = v$  is needed:

$$\text{Recall } I(B_Z(t); x, w) = \frac{E(e^{-B_Z(t)Z} | X=x, W=w)}{E(e^{-B_Z(t)Z} | X=x, W=w)},$$

## Case 1: Continuous & Normal

Assume  $Z \mid V \sim N(\mu(v, \theta), \sigma^2)$ .

- Use MGF of normal variables:  
 $M_Z(s) = E(e^{zs}) = e^{\mu s + \frac{1}{2}\sigma^2 s^2}$  and  
 $M'_Z(s) = E(Ze^{zs}) = (\mu + \sigma^2 s)e^{\mu s + \frac{1}{2}\sigma^2 s^2}$
- $I(B_Z(t); v) = \mu(v, \theta) - \sigma^2 B_Z(t)$

## Case 2: Binary Variable

Assume  $Z \mid V \sim$  logistic regression with  $E(Z \mid V = v) = \text{expit}(\mu(v, \theta))$ .

- $E(Ze^{zs}) = 1 \cdot e^s \cdot P(Z = 1)$  and  
 $E(e^{zs}) = e^0 P(Z = 0) + e^s P(Z = 1)$ .
- $I(B_Z(t); v) = \text{expit}(\mu(x) - B_Z(t))$

*Hypothesis Test then:  $H(t;0)=H(t;1)$*

The collapsability property of the Aalen additive hazards model allows for

- the direct exposure effects and indirect exposure effects to be relatively easily separated,
- the G-computation to be applicable to adjust for time-varying confounding,
- the simulation under marginal structural additive hazards models,  $\lambda(t \mid \hat{x} = x) = \beta_0(t) + \beta_X(t)x$ , to be done by simulating from the additive hazard model,  $\lambda(T \mid x, z)$ .

## Discussion - Exception: the Frailty Model I

Simulating from the Cox model is not as straightforward as described above, since we have that non-linearity leads to  $\mathcal{T}(t) \neq \beta_X$ . However, there is an exception for generating data from a marginal structural Cox model.

Consider the following frailty model for the multiplicative conditional hazard model,

$$\lambda(t \mid x, z) = z\lambda^*(t \mid x).$$

Then, define  $\phi_Z(\Lambda^*(t \mid x)) = E(\exp(-Z\Lambda^*(t \mid x)))$  as the Laplace transform for computing the population-averaged hazard rate

$$\lambda(t \mid \hat{x} = x) = -\frac{d}{dt} \log(\phi_Z(\Lambda^*(t \mid x))).$$

It follows that the cumulative conditional hazard is

$$\Lambda^*(t \mid x) = \phi_Z^{-1}(\exp[-\Lambda(t \mid \hat{x} = x)]).$$

## Discussion - Exception: the Frailty Model II

By choosing  $Z \sim \Gamma(1/\theta, \theta)$ , we have  $\phi_Z(v) = (1 + \theta v)^{-\frac{1}{\theta}}$ . Hence, the Laplace transform of  $\Lambda^*(t | x)$  is

$$\begin{aligned}\phi_Z(\Lambda^*(t | x)) &= \exp(-\Lambda(t | \hat{x} = x)) \\ &= (1 + \theta \Lambda^*(t | x))^{-\frac{1}{\theta}}.\end{aligned}$$

To isolating the cumulative hazard, we perform

$$\begin{aligned}\exp[(-\theta)(-\Lambda(t | \hat{x} = x))] &= (1 + \theta \Lambda^*(t | x))^{-\frac{1}{\theta}(-\theta)} \\ \frac{\exp[\theta \Lambda(t | \hat{x} = x)] - 1}{\theta} &= \Lambda^*(t | x).\end{aligned}$$

Finally, the hazard rate is derived as

$$\frac{d}{dt} \Lambda^*(t | x) = \exp[\theta \Lambda(t | \hat{x} = x)] \lambda(t | \hat{x} = x) = \lambda^*(t | x)$$

## Discussion - Exception: the Frailty Model III

We want the population-averaged causal hazard function to be under the multiplicative model, that is

$$\lambda(t \mid \hat{x} = x) = \lambda_0(t) \exp(\beta_X),$$

and consequently,

$$\Lambda(t \mid \hat{x} = x) = \Lambda_0(t) \exp(\beta_X),$$

the hazard rate is

$$\lambda^*(t \mid x) = \exp[\theta \Lambda_0(t) \exp(\beta_X)] \lambda_0(t) \exp(\beta_X).$$

Finally, the frailty model for the hazard function can be rewritten as

$$\lambda(t \mid x, z) = z \exp[\theta \Lambda_0(t) \exp(\beta_X)] \lambda_0(t) \exp(\beta_X).$$

Therefore, by generating  $Z \sim \Gamma(1/\theta, \theta)$  and  $T$  under the frailty model, we can successfully perform simulations under this conditional model and obtain the population-averaged causal effects

# Discussion - Pros&Cons: G-computation

G-computation estimates the amount of confounding bias.

- Pros:

- More stable estimators compared to alternative techniques aiming, for example, at modeling the exposure like Inverse Probability Weighted,  $Weight = 1/P(\text{receiving treatment} \mid \text{health status})$ . This method can indeed generate high variability.
- The standard models it relies on can be validated using well-known techniques, such as Schoenfeld residuals or their scaled version.

- Cons:

- Extrapolation is more likely to happen.

# Discussion - Continuous Case I

Let us define  $X$  to be continuous and assume the Aalen additive hazards model.

Recall the causal exposure effect is

$$\beta_X(t) = \lambda(t \mid \hat{X} = x + 1) - \lambda(t \mid \hat{X} = x).$$

Whereas, if we subtract the observed hazards rates at dose  $x + 1$  and  $x$ , we end up with the exposure effect and the confounding bias, that is  $\lambda(t \mid \hat{X} = x + 1) - \lambda(t \mid \hat{X} = x)$ .

Hence, the confounding bias formula on the hazard scale is

$$-\beta_Z(t) [I(x + 1, \beta_Z(t)) - I(x, \beta_Z(t))],$$

while on the cumulative hazard scale is

$$\int_0^t I(x + 1, \beta_Z(t)) - I(x, \beta_Z(t)) \frac{d}{d\beta_Z(t)}.$$

An estimate can be obtained by plugging-in  $\hat{\beta}_Z(t)$ , the estimate from the additive hazards model.

# Discussion - Continuous Case II

Depending on the distribution of  $Z$ , the confounding bias is computed in different ways.

- $Z \mid X = x \sim N(\mu(x), \sigma^2)$

$$l(x, \beta_Z(t)) = \mu(x) - \sigma^2 \beta_Z(t)$$

- $Z$  is binary with  $E[Z \mid X = x] = \text{expit}(\mu(x))$ , where  $\text{expit}(a) = \frac{e^a}{1+e^a}$

$$l(x, \beta_Z(t)) = \text{expit}(\mu(x) - \beta_Z(t))$$

# Discussion - Multidimensional Z

If  $Z$  is multidimensional, we can model it non-parametrically.

Let

$$l(x, \beta_Z(t)) = \frac{\sum_i Z_i \exp(-\beta_Z(t) Z_i) W_X(x_i)}{\sum_i \exp(-\beta_Z(t) Z_i) W_X(x_i)},$$

where  $W_X(x)$  are weights assigned to each individual.

The k-nearest neighbor method would assign a high weight if the observed exposure value is close to a given target value, and a low weight if the observed value differs from the target value.

# The End

Thank you for your attention!